Adding, Subtracting, and Multiplying Complex Numbers

Adding and subtracting complex numbers, multiplying them, simplifying powers of the imaginary unit, and building or checking complex solutions. Write every answer in standard form $a + bi$ with a and b real.

- $i = \sqrt{-1}$, so $i^2 = -1$. That single substitution is what turns a product back into $a + bi$.
- Add and subtract by combining real parts with real parts and imaginary parts with imaginary parts — exactly like combining like terms.
- Multiply by distributing (FOIL), then replace every i^2 with -1 and re-collect.
- Powers of i cycle with period 4: $i, -1, -i, 1$. Divide the exponent by 4 and keep the remainder.

1. Simplify: $(3 + 5i) + (4 - 2i)$

Answer: ________________

2. Simplify: $(8 - i) - (3 + 6i)$

Answer: ________________

3. Multiply and write in standard form: $3i(4 - 7i)$

Answer: ________________

4. Multiply and write in standard form: $(2+3i)(5-i)$

Answer: ________________

5. Simplify i^{18}. State the remainder you used when you divided the exponent by 4.

Answer: ________________

6. Simplify: $(-4+7i)+(9-7i)-(2-3i)$

Say in one sentence what happened to the imaginary parts of the first two numbers.

Answer: ________________

7. Multiply $(5+2i)(5-2i)$ and write the result in standard form. Then say in one sentence why the answer has no imaginary part.

Answer: ________________

8. Expand $(4-3i)^2$ and write the result in standard form. Remember that squaring a binomial means multiplying it by itself.

Answer: ________________

9. Simplify: $i^{43} + i^{20}$

Answer: ________________

10. Solve $x^2 - 6x + 13 = 0$ over the complex numbers. Give both solutions in standard form.

Answer: ________________

11. Multiply the three factors and write the result in standard form: $(2+i)(3-2i)(1+i)$
Multiply the first two factors, simplify to standard form, and only then multiply by the third.

Answer: ________________

12. Synthesis challenge. Multiply out

$$\big(x-(5+i)\big)\big(x-(5-i)\big)$$

and write the product as a quadratic in standard form. Then state the two zeros of that quadratic, and say in one sentence why the product came out with real coefficients.

Answer: ____________________